

Consistency of LLMs in Multiple-Choice Question Answering

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1 Introduction

2 Methodology

3 Results

4 Discussion & Conclusion

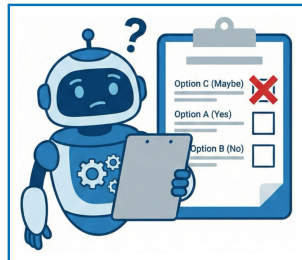
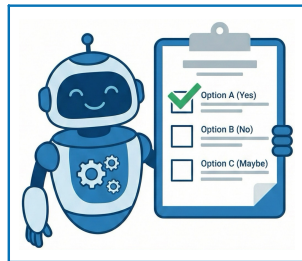
5 Appendix

Context

- LLMs are increasingly used to simulate human respondents [11, 1].
- Assumption: Models “understand” **semantic meaning**.

The Problem

- LLMs are probabilistic text generators, not survey takers.
- Mechanical Instability** [13]: Model opinions can change if trivial changes are made.



AI generated figure

Outline

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Family	Specific Model Name	Type
Llama [5]	Llama-3.1-8B	Base
	Llama-3.1-8B-Instruct	Instruction-Tuned
Mistral [6]	Mistral-7B-v0.3	Base
	Mistral-7B-Instruct-v0.3	Instruction-Tuned
Qwen [12]	Qwen2.5-7B	Base
	Qwen2.5-7B-Instruct	Instruction-Tuned
Gemma [4]	gemma-2-9b	Base
	gemma-2-9b-it	Instruction-Tuned

Table: Overview of the evaluated models (4 Families $\times$ 2 Variants).

Dataset Overview

- **Source:** Pew Research Center (American Trends Panel) [10].
- **Scope:** $\approx 1,500$ subjective questions (1,235 retained).
- **Topics:** Politics, Social Issues, Economics.

Example Question

"How often do you come across made-up news and information that is intended to mislead the public?"

- A) *Often*
- B) *Sometimes*
- C) *Hardly ever*
- D) *Never*

1. The Reworded Dataset

- **Variations:**
 - Option wording
 - Number of options (4 – 6)
- **Goal:** Measure consistency when options change.

2. The Shuffled Dataset

- Randomized **permutations** of options.
- Identical Content, only position changes.
- **Goal:** Measure consistency when order changes.

Example (Reworded Dataset)

- Very likely → Somewhat likely → Not too likely → Not at all likely
- Almost certainly → Probable → Neutral → Unlikely → Very unlikely

Prompt 1: Numerical

Question...

- 1) Very likely
- 2) Somewhat likely

...

Prompt 2: Alphabetical

Question...

- A) Very likely
- B) Somewhat likely

...

Prompt 3: No Labels

Question...

- Very likely
- Somewhat likely

...

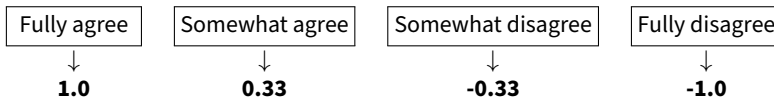
Prompt 4: Extended

Question...

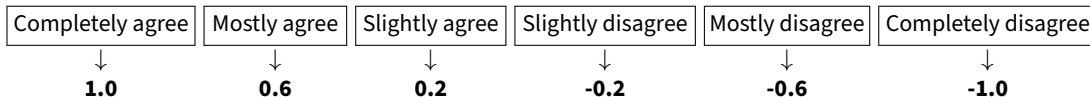
- 1) Very likely
- ...
- 5) Don't know
- 6) Refused

To compare answers all options are mapped to a continuous interval $[-1, 1]$.

1. Standard 4-Point Scale



2. 6-Point Scale



The **Numeric Values** form the basis for all consistency metrics and statistical evaluations.

Question	Options	Model Answer	Numeric Value
Q1	Variation 1	"Fully Agree"	1.0
Q1	Variation 2	"Strongly Agree"	1.0
Q1	Variation 3	"Mostly Agree"	0.6
Q1	Variation 4	"Absolutely agree"	1.0
Q1	Variation 5	"Partly agree"	0.3
Q2	Variation 1	"Often"	1.0
⋮	⋮	⋮	⋮

Let $V = \{v_1, \dots, v_n\}$ be the set of numeric answer values for a single question.

1. Average Pairwise Distance (APD)

$$APD = \frac{1}{K} \sum_{i=1}^n \sum_{j=i+1}^n |v_i - v_j|$$

- $K = \frac{n(n-1)}{2}$ (Total unique pairs).
- Measures **average disagreement**.
- 0.0 indicates perfect consistency.

2. Mean Standard Deviation (MSD)

For each question $j \in \{1, \dots, m\}$:

$$\sigma_{q_j} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (v_{ij} - \bar{v}_j)^2}$$

$$MSD = \frac{1}{m} \sum_{j=1}^m \sigma_{q_j}$$

- Quantifies **volatility** of answers.

Outline

1 Introduction

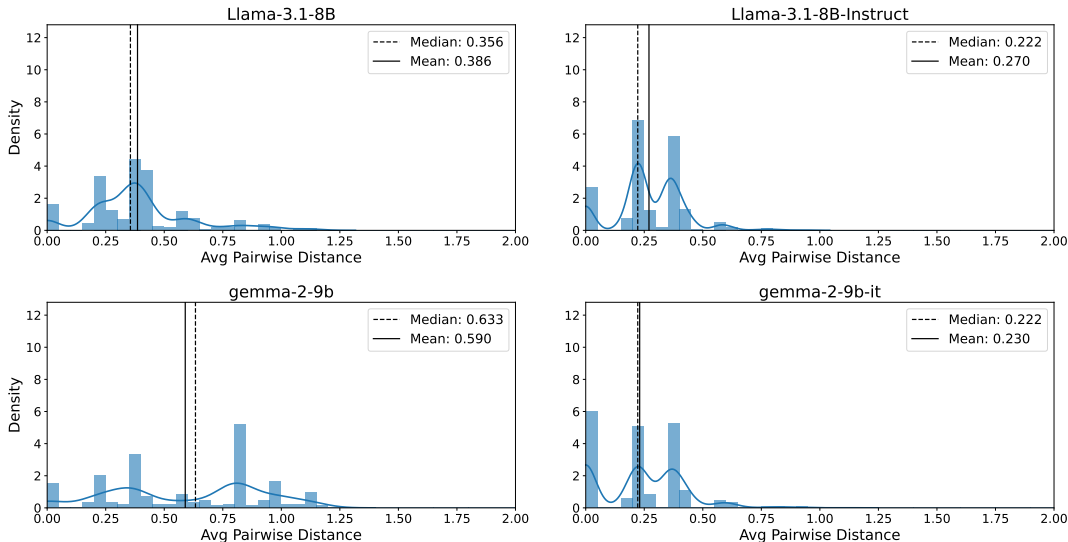
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APD Distribution on Shuffled Dataset (Prompt 1)



Base Models

Reworded Dataset		
Model	P2	P3
Llama	0.315	0.411
Mistral	0.254	0.453
Qwen	0.304	0.415
Gemma	0.325	0.421

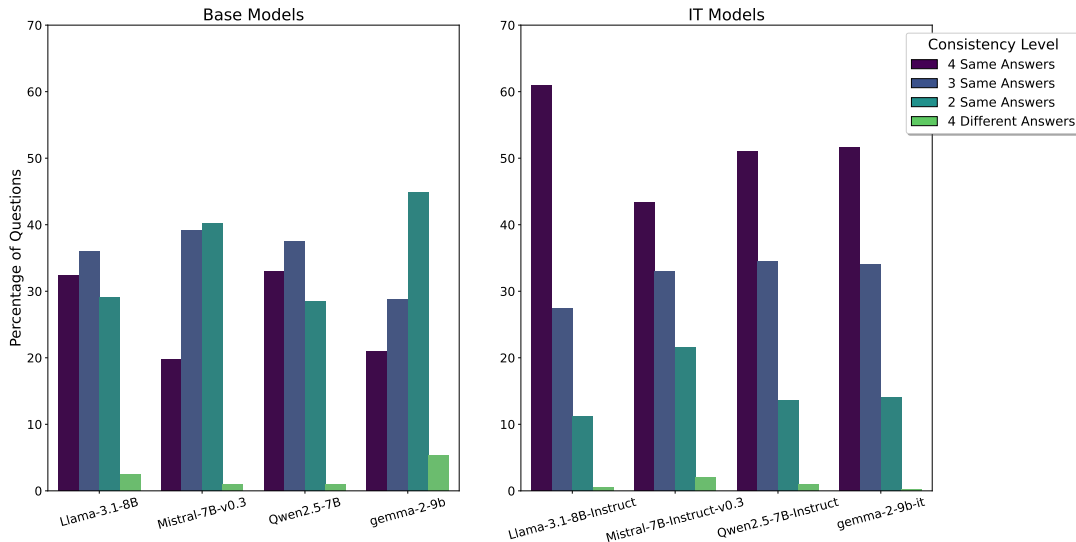
Shuffled Dataset		
Model	P2	P3
Llama	0.403	0.445
Mistral	0.403	0.365
Qwen	0.361	0.363
Gemma	0.346	0.415

Instruction-Tuned Models

Reworded Dataset		
Model	P2	P3
Llama-IT	0.229	0.248
Mistral-IT	0.227	0.287
Qwen-IT	0.249	0.244
Gemma-IT	0.172	0.178

Shuffled Dataset		
Model	P2	P3
Llama-IT	0.233	0.146
Mistral-IT	0.320	0.240
Qwen-IT	0.277	0.206
Gemma-IT	0.187	0.121

Cross-Prompt Consistency Comparison on Reworded Dataset



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IT Models are more consistent

- Instruction Tuning acts as a **reliability filter**, models prioritize semantic meaning [9].
- **Convergence**: Minimizes differences, aligns models to similar standard.

Critique

- **Survival Bias**: Excluding questions with responses that do not map to $[-1, 1]$.
- **Scale Mismatch**: Varying option counts makes perfect numeric consistency (0.0) impossible for intermediate values.

Extensions

- **Question Rewording** [3].
- Investigating **Scale Nuances** (e.g. inclusion of neutral points) [8].

1. Summary of Findings

- **Unstable Base Models:** Responses driven by presentation rather than understanding [14].
- **IT Models are Robust:** Tuning reduces volatility and equalizes consistency.

2. Implications for Synthetic Surveys

- **Model Selection:** Research should rely on Instruction-Tuned variants.
- Single queries are insufficient. Use **Ensemble Prompting** to filter out noise [7].

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This thesis evaluates the consistency of LLMs across two dimensions:
Answer Scale and **Option Order**.

Q1 Impact of Option Wording

To what extent does the rewording of answer options, while maintaining semantic equivalence, impact the consistency of LLM responses in subjective MCQA tasks?

Q2 Impact of Option Order

How does the permutation of answer option order influence the stability of model decisions, and does it reveal positional bias?

Q3 Impact of Prompt Formulation

How does variation in the prompt structure (e.g., labeling style) influence the consistency of LLM answers?

The distribution of answer scales (Cardinality & Polarity) differs by dataset design.

Reworded Dataset ($N = 6,175$)

Scale Type	Count	Pct.
4-point Unipolar	1,858	30.1 %
4-point Bipolar	1,760	28.5 %
5-point Bipolar	1,375	22.3 %
5-point Unipolar	635	10.3 %
6-point Bipolar	512	8.3 %
6-point Unipolar	35	0.6 %

Contains wider variety (incl. 6-point) to test robustness to cardinality changes.

Shuffled Dataset ($N = 7,410$)

Scale Type	Count	Pct.
4-point Unipolar	3,408	46.0 %
4-point Bipolar	2,424	32.7 %
5-point Bipolar	1,350	18.2 %
5-point Unipolar	228	3.1 %

Adheres strictly to original survey options (mostly 4-point scales).

Content Dimension	Count	Percentage
Agreement	404	32.7 %
Quantity	214	17.3 %
Better / Worse	101	8.2 %
Importance	91	7.4 %
Likelihood	85	6.9 %
Positive / Negative	67	5.4 %
Good / Bad	50	4.0 %
Magnitude of Problem	48	3.9 %
Frequency	38	3.1 %
Performance (“How Well”)	38	3.1 %
Priority	35	2.8 %
Acceptance	31	2.5 %
<i>Others (Reason, Concern, etc.)</i>	33	2.7 %
Total	1,235	100.0 %

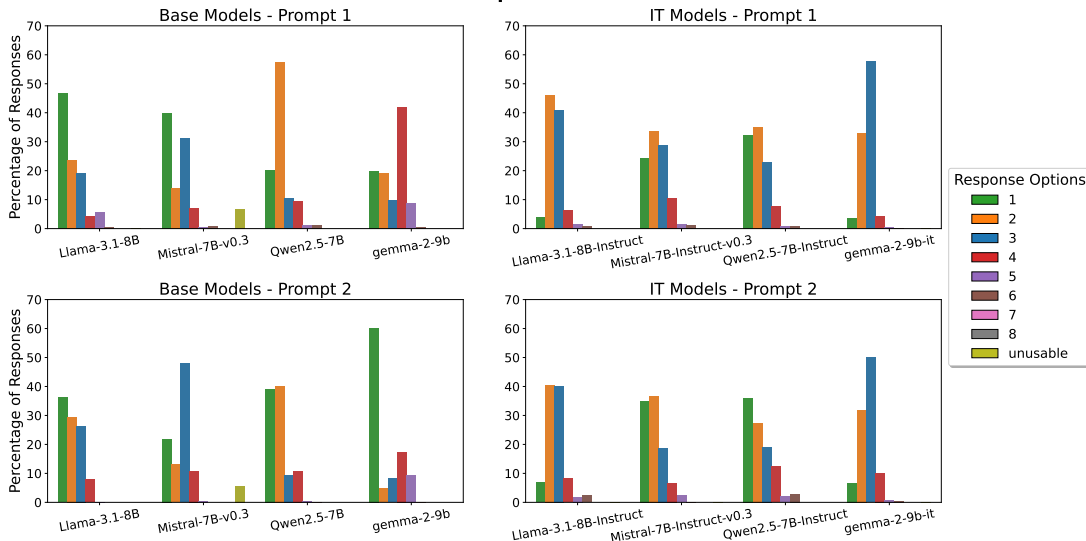
Appendix: Survival Bias (Percentage of Excluded Questions)

Family	Base Models		IT Models	
	Unusable	P4 Refusal	Unusable %	P4 Refusal
	(All Prompts)	(Ext. Option)	(All Prompts)	(Ext. Option)
Llama	< 0.1 %	3.5 %	< 0.1 %	9.7 %
Mistral	17.8 %	2.0 %	0.3 %	21.8 %
Qwen	< 0.1 %	4.3 %	< 0.1 %	14.4 %
Gemma	0.0 %	0.1 %	0.1 %	43.4 %

Average exclusion rates across Reworded and Shuffled datasets.

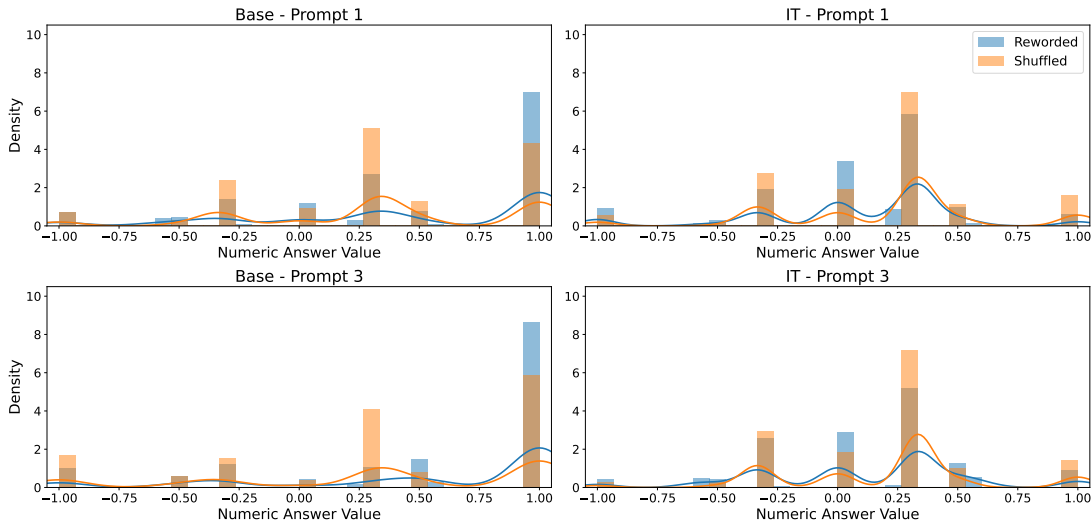
Appendix: Instruction Tuning Eliminates Mechanical Instability

Distribution of Responses on Reworded Dataset

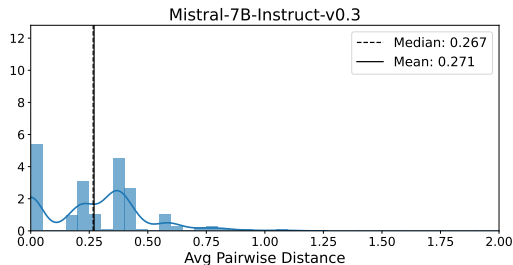
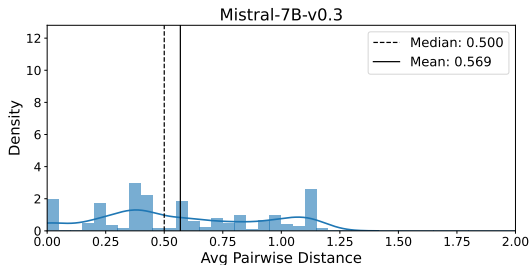
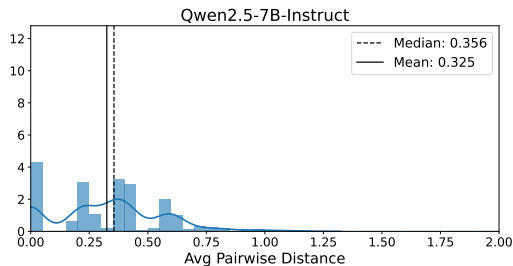
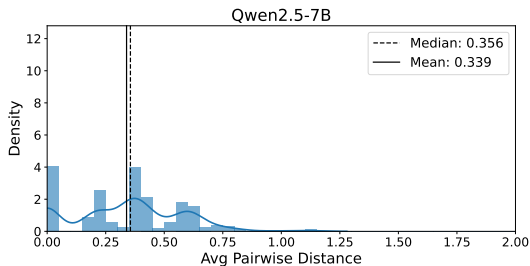


Appendix: Overlapping Histograms (Reworded vs. Shuffled)

Llama: Answer Distribution Base vs IT (Prompts 1 & 3)



APD Distribution on Shuffled Dataset (Prompt 1)



Base Models

Reworded Dataset

Model	P1	P2	P3	P4
Llama	0.357	0.315	0.411	0.340
Mistral	0.302	0.254	0.453	0.294
Qwen	0.254	0.304	0.415	0.224
Gemma	0.344	0.325	0.421	0.229

Shuffled Dataset

Model	P1	P2	P3	P4
Llama	0.367	0.403	0.445	0.328
Mistral	0.508	0.403	0.365	0.510
Qwen	0.323	0.361	0.363	0.330
Gemma	0.553	0.346	0.415	0.378

Instruction-Tuned Models

Reworded Dataset

Model	P1	P2	P3	P4
Llama-IT	0.207	0.229	0.248	0.203
Mistral-IT	0.221	0.227	0.287	0.206
Qwen-IT	0.232	0.249	0.244	0.241
Gemma-IT	0.174	0.172	0.178	0.164

Shuffled Dataset

Model	P1	P2	P3	P4
Llama-IT	0.277	0.233	0.146	0.274
Mistral-IT	0.263	0.320	0.240	0.249
Qwen-IT	0.310	0.277	0.206	0.273
Gemma-IT	0.232	0.187	0.121	0.220

Let $V = \{v_1, \dots, v_n\}$ be the set of numeric answer values for a single question.

3. Maximum Pairwise Distance (MPD)

$$MPD = \max_{1 \leq i < j \leq n} |v_i - v_j|$$

- Captures the **worst-case divergence**.
- Switch from 1 to $-1 \rightarrow MPD = 2$.

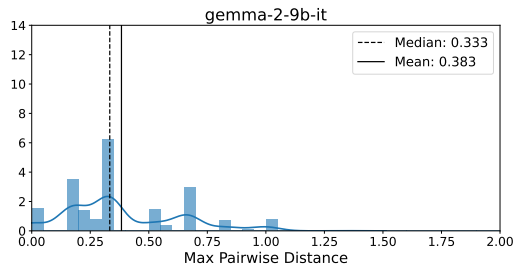
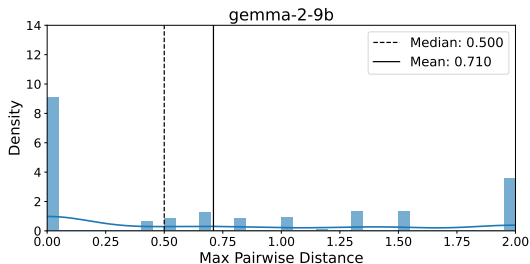
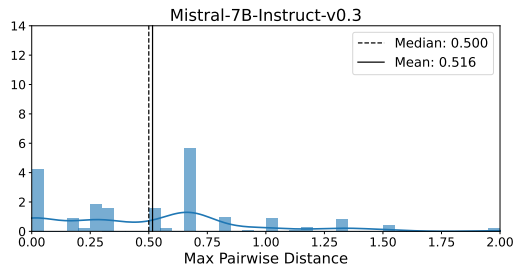
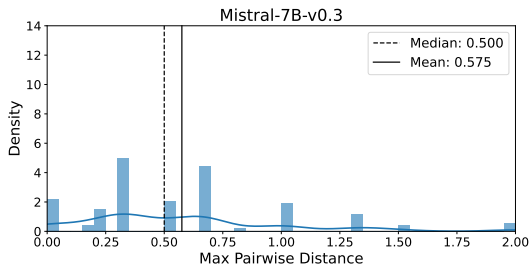
Let X, Y be the sets of numeric answers for two different prompts.

4. Pearson Correlation (r)

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}$$

- Evaluates **consistency across different prompt variations**.
- High (> 0.7): Preserves semantic ordering.
- Low (< 0.4): Changes alter perception.

MPD Distribution on Reworded Dataset (Prompt 2)



Base Models

Model	1-2	1-3	1-4	2-3	2-4	3-4
Llama	0.580	0.398	0.644	0.464	0.536	0.461
Mistral	0.485	0.408	0.454	0.354	0.336	0.278
Qwen	0.608	0.575	0.750	0.467	0.560	0.481
Gemma	0.534	0.374	0.564	0.473	0.627	0.504

Instruction-Tuned Models

Model	1-2	1-3	1-4	2-3	2-4	3-4
Llama-IT	0.820	0.701	0.894	0.712	0.797	0.683
Mistral-IT	0.707	0.722	0.913	0.646	0.709	0.703
Qwen-IT	0.818	0.743	0.883	0.737	0.799	0.726
Gemma-IT	0.860	0.778	0.897	0.820	0.841	0.749

Base Models

Model	1-2	1-3	1-4	2-3	2-4	3-4
Llama	0.545	0.354	0.594	0.441	0.511	0.413
Mistral	0.309	0.183	0.496	0.191	0.269	0.244
Qwen	0.608	0.428	0.849	0.431	0.572	0.400
Gemma	0.468	0.155	0.499	0.312	0.557	0.378

Instruction-Tuned Models

Model	1-2	1-3	1-4	2-3	2-4	3-4
Llama-IT	0.797	0.636	0.796	0.686	0.778	0.644
Mistral-IT	0.730	0.709	0.901	0.627	0.728	0.718
Qwen-IT	0.886	0.783	0.921	0.777	0.882	0.780
Gemma-IT	0.849	0.733	0.856	0.784	0.824	0.717

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